

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 19

DATE : 12-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 19

Topics

DATE : 12 - 01 - 2020**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS** : Physical World & Measurement**CHEMISTRY** : POC + Elemental Analysis + Polymer, Co-ordination Compound**ZOOLOGY** : Evolution + Biomolecules

SPACE FOR ROUGH WORK

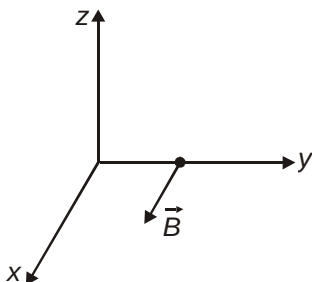
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|---|--|
| <p>1. Red light differs from blue light in its
 (1) Speed (2) Frequency
 (3) Intensity (4) Amplitude</p> <p>2. Which has the largest wavelength?
 (1) Radio wave (2) X-ray
 (3) Ultraviolet ray (4) Infra-red ray</p> <p>3. In an electromagnetic wave, the amplitude of electric field is 1 V/m. What is average energy density of electric field?
 (1) $2.2 \times 10^{-12} \text{ J/m}^3$
 (2) $4.4 \times 10^{-12} \text{ J/m}^3$
 (3) $3 \times 10^{-9} \text{ J/m}^3$
 (4) $12 \times 10^{-12} \text{ J/m}^3$</p> <p>4. Electromagnetic wave is deflected by
 (1) Electric field
 (2) Magnetic field
 (3) Both of (1) & (2)
 (4) Neither electric field nor magnetic field</p> <p>5. The speed of electromagnetic waves depends upon
 (1) Electric field
 (2) Magnetic field
 (3) Intensity
 (4) Medium, in which it travels</p> <p>6. Refractive index of a medium is given by
 (1) $\sqrt{\mu_r \epsilon_r}$ (2) $\sqrt{\mu_0 \epsilon_0}$
 (3) $\frac{1}{\sqrt{\mu_r \epsilon_r}}$ (4) $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$</p> | <p>7. Displacement current is set up between the plates of the capacitor when the potential difference across the plates is
 (1) Maximum (2) Zero
 (3) Minimum (4) Varying</p> <p>8. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic field vectors of the electromagnetic waves, then the direction of propagation of the electromagnetic waves is that of
 (1) $\vec{E}$ (2) $\vec{B}$
 (3) $\vec{E} \times \vec{B}$ (4) $\vec{B} \times \vec{E}$</p> <p>9. Which of the following pairs of space and time varying electric and magnetic field $\vec{E} \times \vec{B}$ would generate a plane electromagnetic wave travelling in y-direction?
 (1) E_x, B_y (2) E_z, E_x
 (3) E_y, B_x (4) E_y, B_z</p> <p>10. If an electromagnetic wave propagating through vacuum is describe by $E_y = E_0 \sin(kx - \omega t)$; $B_z = B_0 \sin(kx - \omega t)$, then
 (1) $E_0 k = B_0 \omega$ (2) $E_0 B_0 = \omega k$
 (3) $E_0 \omega = B_0 k$ (4) $E_0 B_0 = \frac{\omega}{k}$</p> <p>11. Out of the following, choose the ray which does not travel with the velocity of light
 (1) X-ray (2) Microwave
 (3) γ-rays (4) β-rays</p> |
|---|--|

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12. In a plane electromagnetic wave, which of the following have zero average value?
- Magnetic field
 - Magnetic energy
 - Electric field
 - Electric energy
- (1) (a), (c) (2) (b), (c)
(3) (a), (d) (4) All of these
13. The ratio of amplitude B_0 and E_0 of the magnetic and electric fields associated with an electromagnetic wave is
- (1) $\frac{1}{c}$ (2) c
(3) c^2 (4) $\frac{1}{c^2}$
14. The figure gives the magnetic field of an electromagnetic wave at a certain point and at a certain instant. The wave transports energy in negative y -direction. Then the direction of electric field at this point and instant is

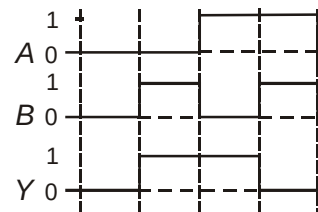


- (1) z -axis (2) $-z$ -axis
(3) x -axis (4) $-x$ -axis

15. The amplitude of electric field at a distance r from a source of power P is (taking 100% efficiency)

(1) $\sqrt{\frac{P}{2\pi r^2 c \epsilon_0}}$ (2) $\sqrt{\frac{P}{4\pi r^2 c \epsilon_0}}$
(3) $\sqrt{\frac{P}{8\pi r^2 c \epsilon_0}}$ (4) $\frac{P}{2\pi r^2 c \epsilon_0}$

16. In a common emitter amplifier the phase difference between input and output voltage is
- (1) Zero (2) $\pi/4$
(3) $\pi/2$ (4) π
17. In which of the configurations of a transistor, the power gain is highest?
- (1) Common base
(2) Common emitter
(3) Common collector
(4) Same in all the three
18. There is a gate with two inputs A and B , and one output Y . Using the input and output wave forms identify the gate



- (1) NAND
(2) NOR
(3) EXCLUSIVE-OR
(4) AND

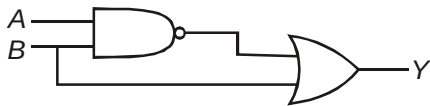
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19. An n -type semi-conductor has donor levels at 500 meV above the valence band, the frequency of light required to create a hole is nearly
- (1) 8×10^{13} Hz
 - (2) 12×10^{13} Hz
 - (3) 22×10^{13} Hz
 - (4) 15×10^{13} Hz

20. In a common emitter amplifier, when a signal of 40 mV is added to the input voltage the base current changes of 100 μ A and emitter current changes by 2.1 mA, then the trans conductance is

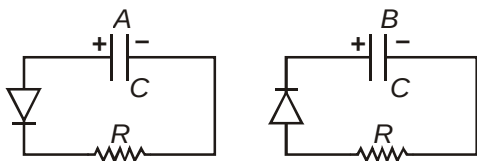
- (1) $\frac{1}{20} \Omega^{-1}$
- (2) $\frac{1}{50} \Omega^{-1}$
- (3) $50 \Omega^{-1}$
- (4) $15 \Omega^{-1}$

21. Give the output of the given logic gate



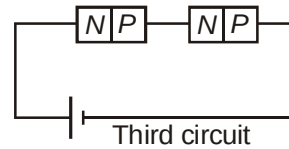
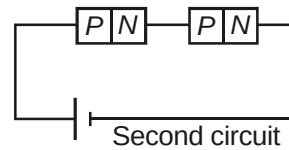
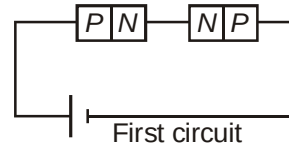
- (1) $Y = \bar{A}B + \bar{B}$
- (2) $Y = \bar{A}B + \bar{B}A$
- (3) $Y = 1$
- (4) $Y = (\bar{A} + \bar{B})\bar{B}$

22. Two identical capacitors A and B are charged to the same potential V and are connected in two circuits at $t = 0$ as shown in the figure. The charges on the capacitors at time $t = RC$ are respectively



- (1) VC, VC
- (2) $VC/e, VC$
- (3) $VC, VC/e$
- (4) $VC/e, VC/e$

23. Two identical P - N junctions may be connected in series with a battery in three ways as shown in the adjoining figure. The potential drop across the P - N junctions are equal in



- (1) First and second circuits
- (2) Second and third circuits
- (3) Third and first circuits
- (4) All of these

24. The relationship between α and β is given by

- (1) $\alpha = \beta$
- (2) $\alpha = \frac{1}{\beta}$
- (3) $\beta = \frac{\alpha}{1-\alpha}$
- (4) $\beta = \frac{\alpha}{1+\alpha}$

25. The material suitable for making a solar cell is

- (1) Pbs
- (2) GaAs
- (3) CdSe
- (4) Ge

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26. A diode can convert

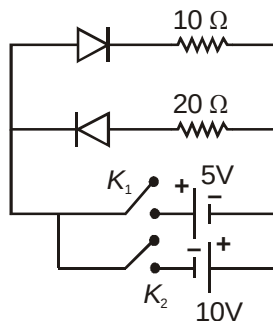
- (1) A.C. into D.C. but not D.C. into A.C.
- (2) D.C. into A.C. but not A.C. into D.C.
- (3) A.C. into D.C. and D.C. into A.C.
- (4) All of these

27. In semiconductor, which of the following relation is correct at thermal equilibrium

- (1) $n_i = n_e = n_n$
- (2) $n_i^2 = n_e n_n$
- (3) $n_e = n_n = 0$

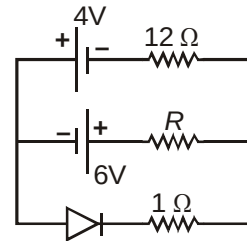
(4) $n_i = \frac{n_e}{n_n}$

28. In the given circuit calculate the ratio of currents through the batteries if (1) key K_1 is pressed, K_2 open and then (2) key K_2 is pressed, key K_1 is opened



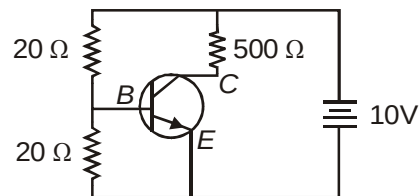
- (1) $\frac{1}{2}$
- (2) 4
- (3) 2
- (4) 1

29. In the given circuit calculate the current through resistance R in the figure ($R = 12 \Omega$)



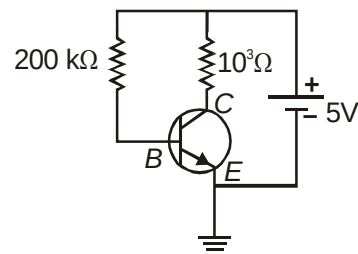
- (1) 0.92 A
- (2) 0.13 A
- (3) 0.72 A
- (4) 0.42 A

30. In the given circuit of the current amplification is 20 and the base current is $50 \times 10^{-6} \text{ A}$ then the potential drop between the collector emitter junction is



- (1) 6 V
- (2) 9.5 V
- (3) 8 V
- (4) 12 V

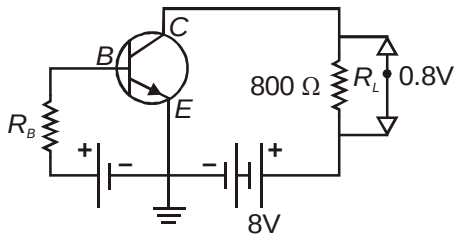
31. In the given network, if $V_{BE} = 1 \text{ V}$ then, the input resistance of the base-emitter junction is



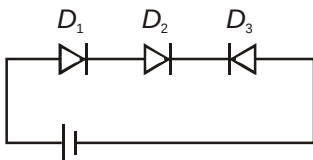
- (1) $15 \times 10^4 \Omega$
- (2) $5 \times 10^4 \Omega$
- (3) $20 \times 10^5 \Omega$
- (4) $4 \times 10^4 \Omega$

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32. If the current amplification factor for a transistor connected in common emitter configuration is 100 and input resistance is $200\ \Omega$, then calculate the power gain

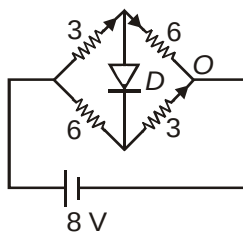


- (1) 4×10^4
(2) 5×10^3
(3) 7×10^3
(4) 45×10^4
33. Three diodes are connected as shown in figure



if diodes are ideal then choose correct statement

- (1) D_1 is forward biased
(2) D_2 is forward biased
(3) D_3 is forward biased
(4) D_3 is reverse biased
34. In figure shown find current through diode

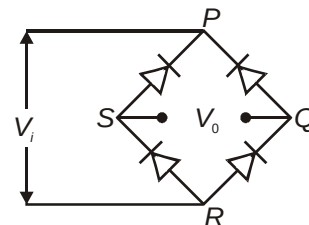


- (1) $\frac{4}{3}\text{ A}$ (2) $\frac{2}{3}\text{ A}$
(3) 2 A (4) Zero

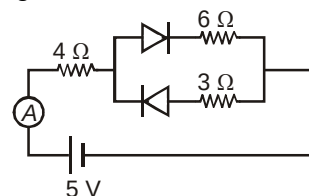
35. Choose the correct statement from following

- (1) p -type semiconductor is positively charged
(2) Phosphor is acceptor type impurity
(3) Donor atom electrons have energy state that is slightly below the lowest energy state of CB(Conduction Band)
(4) None of these

36. In diagram, the input is across the terminal P and R and the output is across S and Q . Then output is



- (1) Half wave rectified
(2) Full wave rectified
(3) Same as the input
(4) Zero
37. Reading of Ammeter in following circuit



- (1) $\frac{5}{6}\text{ A}$ (2) $\frac{5}{7}\text{ A}$
(3) $\frac{1}{2}\text{ A}$ (4) $\frac{5}{4}\text{ A}$

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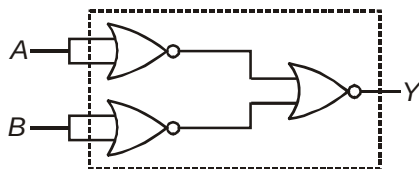
38. 'A + AB' Boolean equation is equal to

- (1) A
- (2) B
- (3) A + B
- (4) AB

39. How many NAND gates are required to make OR gate?

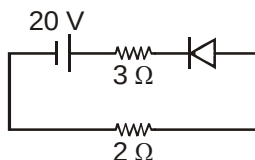
- (1) 1
- (2) 2
- (3) 3
- (4) 4

40. The combination of NOR gates shown here are equivalent to



- (1) OR
- (2) AND
- (3) NAND
- (4) XOR

41. In the given circuit diode is ideal. The potential difference across $2\ \Omega$ resistance is



- (1) 10 V
- (2) 8 V
- (3) 20 V
- (4) Zero

42. In common base amplifier, the phase difference between the input signal voltage and output voltage is

- (1) $\frac{\pi}{4}$
- (2) π
- (3) Zero
- (4) $\frac{\pi}{2}$

43. In *p-n-p* transistor the collector current is 8 mA. If 80% of holes emitted reach the collector, the emitter current will be

- (1) 9 mA
- (2) 11 mA
- (3) 10 mA
- (4) 2 mA

44. In a common emitter amplifier, input resistance is $3\ \Omega$ and load resistance is $24\ \Omega$. What is voltage gain? (Take $\alpha = 0.8$)

- (1) 16
- (2) 24
- (3) 32
- (4) 8

45. A potential Barrier of 0.2 V exists across a *p-n* junction. A uniform electric field of magnitude $10^6\ \text{V/m}$ exists in depletion region. The width of depletion region is

- (1) $4 \times 10^{-7}\ \text{m}$
- (2) 0.1 mm
- (3) $2 \times 10^{-7}\ \text{m}$
- (4) 20 μm

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46. In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{SCN})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$ more stability is shown by
(1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{SCN})_6]^{3-}$
(3) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (4) $[\text{FeCl}_6]^{3-}$
47. One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of AgCl (s). The structure of the complex is
(1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$
(2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \text{Cl} \cdot \text{NH}_3$
(3) $[\text{Co}(\text{NH}_3)_4\text{Cl}] \text{Cl}_2 \cdot \text{NH}_3$
(4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
48. In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is –
(1) 0 (2) +1 (3) +2 (4) –1
49. The number of 3d-electrons remained in Fe^{2+} (At.no. of Fe = 26) ion is –
(1) 4 (2) 5 (3) 6 (4) 3
50. Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :-
(1) In acidic solutions hydration protects copper ions
(2) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
(3) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
(4) Copper hydroxide is an amphoteric substance
51. Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN^- ion towards metal species is :-
(1) c, a (2) b, c
(3) a, b (4) a, b, c
52. The coordination number of a central metal atom in a complex is determined by :-
(1) The number of ligands around a metal ion bonded by sigma and pi-bonds both
(2) The number of ligands around a metal ion bonded by pi-bonds
(3) The number of ligands around a metal ion bonded by sigma bonds
(4) The number of only anionic ligands bonded to the metal ion
53. Which one of the following complexes is an outer orbital complex :-
(1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$
(3) $[\text{Fe}(\text{CN})_6]^{4-}$ (4) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
(Atomic nos.: Mn=25 ; Fe=26 ; Co=27 ; Ni = 28)
54. Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ?
(1) Cyanocobalamin is vitamin B_{12} and contains cobalt
(2) Haemoglobin is the red pigment of blood and contains iron
(3) Chlorophylls are green pigments in plants and contain calcium
(4) Carboxypeptidase - A is an enzyme and contains zinc

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55. The correct order of magnetic moments (spin only values in B.M.) among is :-
 (1) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 (2) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
 (4) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (Atomic no. : Mn = 25, Fe = 26, Co = 27)
56. For octahedral complex, the value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is
 (1) d^4 (in strong ligand field)
 (2) d^4 (in weak ligand field)
 (3) d^3 (in weak as well as in strong field)
 (4) d^5 (in strong ligand field)
57. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is
 (1) pentaammine nitrito-N- cobalt (II) chloride
 (2) pentaammine nitrito-N- cobalt (III) chloride
 (3) nitrito-N- pentaamminecobalt (III) chloride
 (4) nitrito-N- pentaamminecobalt (II) chloride
58. Nickel ($Z=28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron in the nickel and geometry of this complex ion are, respectively.
 (1) one, square planar
 (2) two, square planar
 (3) one, tetrahedral
 (4) two, tetrahedral
59. How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion ?
 (1) One (2) Two
 (3) Six (4) Three
60. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. Ni= 28)
 (1) 0 (2) 1.73
 (3) 2.84 (4) 4.90
61. Which one of the following has a square planar geometry :- (Co = 27, Ni = 28, Fe=26, Pt = 78)
 (1) $[\text{CoCl}_4]^{2-}$
 (2) $[\text{FeCl}_4]^{2-}$
 (3) $[\text{NiCl}_4]^{2-}$
 (4) $[\text{PtCl}_4]^{2-}$
62. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4^{2-})] \text{NO}_2^-$ (where en is ethylene diamine) are, respectively
 (1) 6 and 2 (2) 4 and 2
 (3) 4 and 3 (4) 6 and 3
63. In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_0 be the highest ?
 (1) $[\text{Co}(\text{CN})_6]^{3-}$
 (2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$

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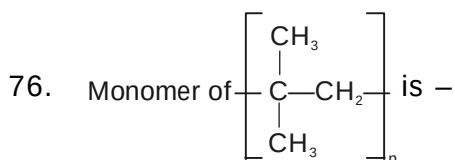
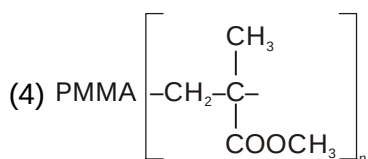
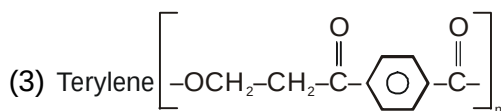
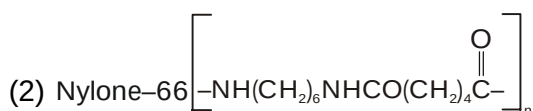
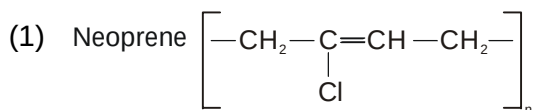
64. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is :-
(At. mass of Ag = 108 u)
(1) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
(2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
(3) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$
(4) $[\text{CoCl}_3(\text{NH}_3)_3]$
65. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong ?
(1) The complex is an outer orbital complex
(2) The complex gives white precipitate with silver nitrate solution
(3) The complex involves d^2sp^3 hybridisation and is octahedral in shape
(4) The complex is paramagnetic
66. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is :-
(1) 2.82 BM (2) 1.41 BM
(3) 1.82 BM (4) 5.46 BM
67. Among the ligands NH_3 , en, CN^- and CO the correct order of their increasing field strength, is :-
(1) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$
(2) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
(3) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$
(4) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
68. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ?
(1) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (2) $[\text{Cr}(\text{en})_3]\text{Br}_3$
(3) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (4) $[\text{Cr}(\text{en})\text{Br}_4]^-$
69. AgCl precipitate dissolves in NH_3 due to the formation of –
(1) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
(2) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
(3) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
(4) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
70. Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$:-
(1) Bromoaquatetraaminecobalt(III) nitrate
(2) Bromoaquatetraaminocobalt(III) nitrate
(3) Bromoaquatetraamminecobalt(III) nitrate
(4) Tetraammineaquabromocobalt(III) nitrate
71. Nessler's reagent is –
(1) K_2HgI_4 (2) $\text{K}_2\text{HgI}_4 + \text{KOH}$
(3) $\text{K}_2\text{HgI}_2 + \text{KOH}$ (4) $\text{K}_2\text{HgI}_4 + \text{Hg}$
72. Oxidation state of Ag in $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ is
(1) +2 (2) –2 (3) 0 (4) +1
73. Out of $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$
(1) All have identical geometry.
(2) All are paramagnetic.
(3) All are diamagnetic.
(4) $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic but $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ are paramagnetic

SPACE FOR ROUGH WORK

74. $\text{CF}_2=\text{CF}_2$ is a monomer of –

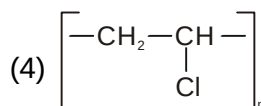
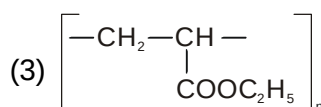
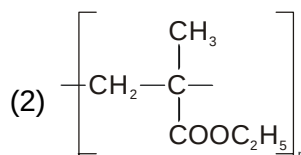
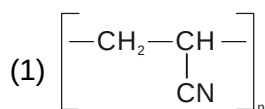
- (1) Teflon (2) Orlon
(3) Polythene (4) Nylon-6

75. Which of the following is not correctly matched –

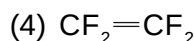
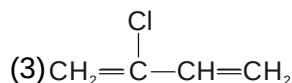
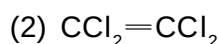
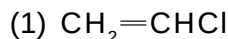


- (1) 2-methyl propene
(2) Styrene
(3) Propylene
(4) Ethene

77. Acrylon is a hard, horny and a high melting material. Which of the following represents its structure –



78. Which one of the following monomers gives the polymer neoprene on polymerization –

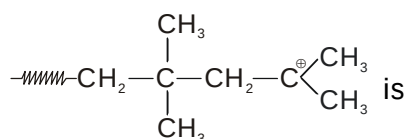


79. Which of the following is a biodegradable polymer

- (1) PHBV
(2) Polythene
(3) Polyvinyl chloride
(4) Nylon-6

SPACE FOR ROUGH WORK

80. The monomer of the polymer –



- (1) $\text{CH}_3\text{CH}=\text{CHCH}_3$
 (2) $\text{CH}_3\text{CH}=\text{CH}_2$
 (3) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$
 (4) $\text{H}_2\text{C}=\text{C}\begin{matrix} \text{CH}_3 \\ \text{CH}_3 \end{matrix}$

81. Which one of the following polymers is prepared by condensation polymerization

- (1) Styrene (2) Nylon-66
 (3) Teflon (4) Rubber

82. Which of the following statements is incorrect?

- (1) Polyethylene contains double bonds
 (2) The monomer used to make teflon is C_2F_4
 (3) PHBV is a biodegradable polymer
 (4) Teflon is used in cooking vessels

83. Melamine polymer is copolymer of

- (1) Melamine and acetaldehyde
 (2) Melamine and formaldehyde
 (3) Phenol and formaldehyde
 (4) None of the above.

84. Polyacrylonitrile, characterized by the repeating unit, is made from which of the following monomer ?

- (1) $\text{CH}_3\text{CH}_2\text{CN}$
 (2) $\text{HOCH}_2\text{CH}_2\text{CH}_3$
 (3) $\text{CH}_3\text{CH}=\text{CHCN}$
 (4) $\text{CH}_2=\text{CHCN}$

85. Bakelite is obtained from phenol and formaldehyde. The initial reaction between the two compounds is an example of

- (1) Aromatic electrophilic substitution
 (2) Aromatic nucleophilic Substitution
 (3) Free radical reaction
 (4) Aldol reaction.

86. On the basis of intermolecular forces, polymers are classified as

- (1) Elastomers, Fibres, Thermoplastics and Thermosetting
 (2) Elastomers, Fibres, Chain growth and Step growth
 (3) Addition polymers and Condensation polymers
 (4) None of these

87. Which of the following pairs is not correctly matched?

- (1) Terylene – condensation polymer of terephthalic acid and ethylene glycol
 (2) Teflon – thermally stable cross linked polymer of phenol and formaldehyde
 (3) Perspex – a homopolymer of methyl methacrylate
 (4) Synthetic rubber – a copolymer of butadiene and styrene

88. 0.3960 gm of an organic compound on combustion gives 0.792 gm CO_2 and 0.324 gm of H_2O . Calculate the % of C and H :-

- (1) 54.55%, 9.09% (2) 9.09%, 54.55%
 (3) 18.55%, 9.09% (4) 9.09%, 18.55%

89. 0.25 gm of an organic compound at NTP gives 31 ml of N_2 gas by Duma's method. Find out of % of N :-

- (1) 18.5% (2) 16.5%
 (3) 15.5% (4) 65.5%

90. 30 ml 0.25 N H_2SO_4 are used in neutralizing NH_3 obtained from 0.75 gm of an organic compound in Kjeldhal's method find out % of N in the compound :-

- (1) 14% (2) 15%
 (3) 16% (4) 28%

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91. Match the column :-

Column-I	Column-II
A. Basic amino acid B. Acidic amino acid C. Neutral amino acid	(i) Alanine (ii) Glycine (iii) Aspartic acid (iv) Glutamic acid (v) Valine (vi) Arginine (vii) Lysine (viii) Phenyl alanine

- (1) A = (i), (ii), (iii); B = (iv), (v)
C = (vi), (vii), (viii)
(2) A = (vi), (vii), (viii); B = (iii), (iv)
C = (i), (ii), (iii)
(3) A = (vi), (vii) ; B = (iii), (iv);
C = (i), (ii), (v), (viii)
(4) A = (v), (vi), (vii); B = (i), (ii), (iii)
C = (iv), (viii)

92. Lipid comes under acid insoluble fraction during analysis of the chemical composition of tissues. Why?

- (1) It has low molecular weight
(2) It has high molecular weight
(3) It is polymer
(4) On grinding, the biomembranes are broken into pieces and form insoluble vesicles

93. The four elements called "Big-four" which make up 95% of all elements found in a living system are

- (1) C, H, O, N (2) C, H, O, P
(3) C, H, O, S (4) C, N, O, P

94. Biomolecules are

- (1) inorganic materials
(2) organic materials
(3) all carbon compound obtained from living tissues
(4) only DNA, RNA

95. Based up on the number of NH_2 and COOH amino acids are

- (1) 3 types
(2) 2 types
(3) 4 types
(4) 20 types

96. The aromatic amino acid (s) is (are)

- (1) Tyrosine
(2) Phenylalanine
(3) tryptophan
(4) All of these

97. Find out the correct match.

- (1) Apoenzyme = Co-factor + Co-enzyme
(2) Conjugated enzyme = Simple enzyme + apoenzyme
(3) Prosthetic group = Co-factor - Co-enzyme
(4) Holoenzyme = Apoenzyme + Co-factor

98. Types of amino acids universally used in protein synthesis are

- (1) 20 (2) 80
(3) 64 (4) 61

99. Which one is the most abundant organic biomolecule in a cell?

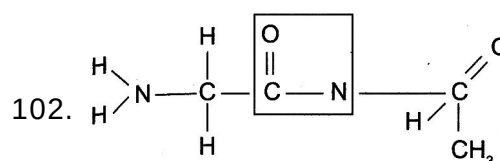
- (1) Water (2) Protein
(3) DNA (4) mRNA

100. Zinc is the cofactor of enzyme

- (1) Hexokinase
(2) Amylase
(3) Carboxypeptidase
(4) Urease

101. Which is the highest structural organization found in all enzymes?

- (1) 2° (2) 3° (3) 1° (4) 4°



What is the structure enclosed by a box?

- (1) Amino acid
(2) Peptide bond
(3) Glycosidic bond
(4) Zwitter ion

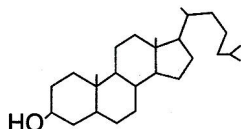
103. Lipids are

- (1) insoluble in water
(2) readily soluble in organic solvent
(3) important constituent of biological membrane
(4) all of these

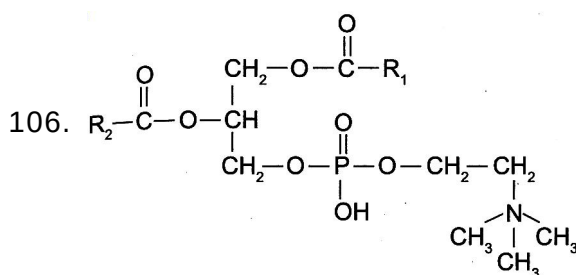
104. The substrate concentration at which the chemical reaction catalyzed by an enzyme attains half of its maximum velocity is termed as

- (1) $\frac{1}{2} V_{\max}$ (2) K_m
(3) $\frac{1}{2} K_m$ (4) $V_{\max}$

105. This molecule is related to



- (1) Phospholipid (2) Lecithin
(3) Cholesterol (4) Oleic acid



The above diagrammatic representation is the formula of

- (1) Lecithin (2) Cholesterol
(3) Uridylic acid (4) Phosphatic acid

107. Which of the following secondary metabolites are used as drugs?

- (1) Abrin and Ricin
(2) Vinblastin and Curcumin
(3) Anthocyanine
(4) Amino acid

108. Three important polysaccharides made up of glucose monomers are

- (1) Sucrose, lactose, and maltose
(2) Chitin, glycogen, and starch
(3) Starch, glycogen, and cellulose
(4) RNA, DNA, and starch

109. The effect of Glucose-6-phosphate on enzyme hexokinase is an example of

- (1) Allosteric modulation
(2) Feed-back inhibition
(3) End product inhibition
(4) All of these

110. Which of the following is the best evidence for the lock and key theory of enzyme action?

- (1) Compounds similar in structure to the substrate inhibit the reaction
(2) Enzymes determine the direction of a reaction
(3) Enzymes are found in living organisms and speed up certain reactions
(4) All isolated enzymes have been identified as proteins

111. (I) Right end of polysaccharide is called reducing end while left end is called non reducing end

(II) Starch can hold I_2 molecules in its helical secondary structure but cellulose being non-helical, cannot hold I_2

(III) Starch and glycogen are branched molecules

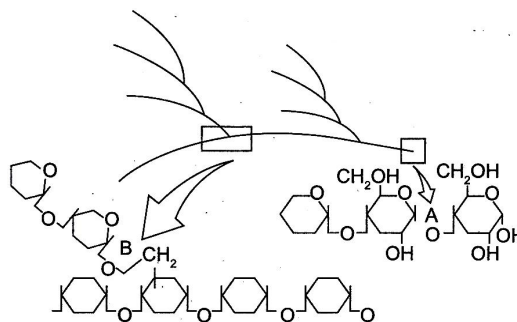
(IV) Starch in plant and glycogen in animals are store houses of energy

- (1) I and IV are correct
(2) II and III are correct
(3) Only IV is correct
(4) All are correct

112. In polysaccharide, the individual monosaccharides are linked by a bond. This bond is formed by dehydration. The bond is

- (1) Ester bond (2) Glycosidic bond
(3) H-bond (4) Ionic bond

113. Identify A and B bonds in the following diagrammatic representation of a portion of glycogen.



- (1) A = 1-4 α -glycosidic bonds, B = 1-6 α -glycosidic bonds
(2) A = 1-6 α -glycosidic bonds, B = 1-4 α -glycosidic bonds
(3) A = 1-1 α -glycosidic bonds, B = 1-1 α -glycosidic bonds
(4) A = 1-4 α -glycosidic bonds, B = 1-4 α -glycosidic bonds

114. Which of the following fatty acids is liquid at room temperature?
(1) Palmitic acid (2) Stearic acid
(3) Oleic acid (4) Arachidic acid
115. Which one of the following statements is wrong?
(1) Sucrose is a disaccharide
(2) Cellulose is a polysaccharide
(3) Uracil is a pyrimidine
(4) Glycine is a sulphur containing amino acid
116. Content of nucleic acid in protoplasm is
(1) 70% (2) 15%
(3) 5% (4) 3%
117. All of the following are found in acid insoluble fraction, except :-
(1) Proteins
(2) Nucleic acid
(3) Polysaccharide
(4) Amino acid
118. If we observe the average composition of cell, then :-
(1) Water is the most abundant constituent
(2) Percentage of carbohydrate is more than lipid
(3) Protein constitute 10–15%
(4) All are correct
119. Which of the following is not a secondary metabolite?
(1) Rubber (2) Carotenoids
(3) Monoterpenes (4) Sugars
120. Which of the following is considered as macromolecule, though they have molecular weight less than 800 daltons?
(1) Proteins (2) Lipids
(3) Polysaccharide (4) Nucleic acid
121. Which of the following polysaccharides contain α 1 – 4 and α 1 – 6 glycosidic linkage?
(A) Starch (B) Glycogen
(C) Chitin (D) Cellulose
(1) A and C (2) A and B
(3) B and C (4) C and D
122. Which of the following polysaccharides is called as animal starch?
(1) Glycogen (2) Inulin
(3) Amylose (4) Chitin
123. Type of linkage present in amylopectin is:-
(1) β 1–4 (2) α 1–4, α 1–6
(3) β 1–6 (4) α 1–4
124. Hydrolysis of maltose yields :-
(1) Glucose + glucose
(2) Fructose only
(3) Galactose + glucose + fructose
(4) Galactose + galactose + glucose + fructose
125. The sweetest naturally occurring sugar is :-
(1) Sucrose (2) Fructose
(3) Maltose (4) Lactose
126. Milk sugar is :-
(1) Lactose (2) Sucrose
(3) Maltose (4) Fructose
127. Glucose is :-
(1) Blood sugar (2) Grape sugar
(3) Dextrose (4) All of these
128. Which of the following is not a function of polysaccharides?
(1) Energy storage
(2) Structural support
(3) Energy source
(4) Storage of genetic information
129. Zwitter ion has :-
(1) Only positive charge
(2) Only negative charge
(3) Either positive or negative charge
(4) Both positive and negative charge
130. Which is the most abundant protein in animal world and in the whole biosphere respectively?
(1) Insulin and Collagen
(2) Albumin and globin
(3) Collagen and RuBisCo
(4) Trypsin and GLUT-4
131. The simplest amino acid is :-
(1) Lysine (2) Glycine
(3) Valine (4) Glutamic acid

132. Which of the following is a sulphur containing amino acid?

- (1) Serine (2) Proline
(3) Cysteine (4) Alanine

133. Which of the following bonds is used for stabilising the primary structure of protein?

- (1) Hydrogen bond
(2) Peptide bond
(3) Hydrophobic interaction
(4) Disulphide bond

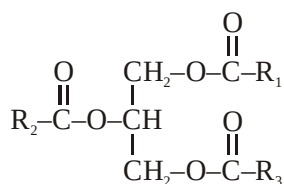
134. A β -pleated sheet organization in a polypeptide is an example of :-

- (1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) Quaternary structure

135. Which of the following statements is not true?

- (1) Galactose is a disaccharide present in milk
(2) Lecithin is a phospholipid present in cell membrane
(3) Lipids are not strictly macromolecules
(4) PUFA are least harmful for heart

136. This is the structure of



- (1) Steroid (2) Triglyceride
(3) Cholesterol (4) Lecithin

137. The biomolecules which provide stability to biological membrane as well as it is precursor of Vitamin-D in animals is :-

- (1) Lecithin (2) Cholesterol
(3) Wax (4) Stearic acid

138. Which one is a disaccharide?

- (1) Galactose (2) Fructose
(3) Maltose (4) Dextrin

139. Carbohydrate is :-

- (1) Polymers of fatty acid
(2) Polymer of amino acids
(3) Polyhydroxy aldehyde or ketone
(4) None

140. True statement for cellulose molecule :-

- (1) β - 1' - 4" linkage. unbranched
(2) β - 1' - 4" linkage. branched
(3) α - 1' - 4" linkage. branched
(4) β - 1' - 6" linkage. unbranched

141. Amylose and Amylopectin chains occur in

- (1) Glycogen (2) Starch
(3) Cellulose (4) Chitin

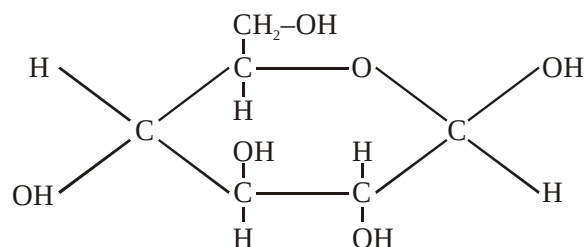
142. One of the major components of cell wall of most fungi is :-

- (1) Chitin (2) Peptidoglycan
(3) Cellulose (4) Hemicellulose

143. A typical fat molecule is made up of :-

- (1) Three glycerol molecules and one fatty acid molecule
(2) One glycerol and three fatty acid molecules
(3) One glycerol and one fatty acid molecule
(4) Three glycerol and three fatty acid molecules

144. Following structure is :-

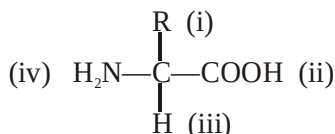


- (1) α -Glucose (2) β -Glucose
(3) Ribose (4) Deoxyribose

145. Which of the following bond is/are found in tertiary structure of protein :-

- (1) peptide bond & hydrogen bond
(2) disulphide & hydrophobic bond
(3) ionic bond
(4) all the above

146. Which of the two groups of the structure of an amino acid given below are involved in peptide bond formation ?



- (1) (i) and (iii) (2) (ii) and (iv)
(3) (iii) and (iv) (4) (i) and (iv)

147. Observe the reaction flow given below :-

- (i) Guanine + X = Guanosine
(ii) Guanosine + Y = Guanylate (or GMP)
Identify X and Y :-

- (1) X = Sugar, Y = Phosphate group
(2) X = Hexose sugar, Y = Phosphate group
(3) X = Ribose sugar, Y = Phosphate group
(4) X = Deoxyribose sugar, Y = Phosphate group

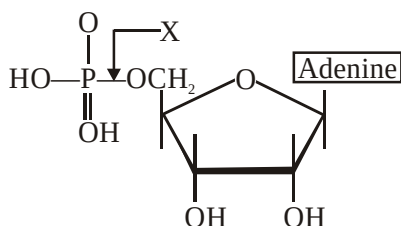
148. Biomolecules are primarily bonded by :-

- (1) Covalent bonds (2) Ionic bonds
(3) Peptide bonds (4) Hydrogen bonds

149. In an experiment, DNA was found to have 31% adenine and 19% guanine. The quantity of cytosine shall be :-

- (1) 38% (2) 31% (3) 19% (4) 62%

150. Which type of bond is indicated by X in the diagram given below ?



- (1) Phosphodiester bond
(2) Ionic bond
(3) Glycosidic bond
(4) Ester bond

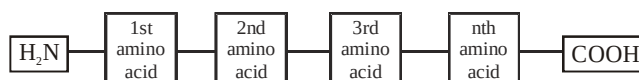
151. Purines are :-

- (1) A, G (2) C, T
(3) U, T (4) A, T

152. The β -pleated sheet protein structure is due to :-

- (1) Coiling of the polypeptide chains
(2) The linking together of two or more polypeptide chains by intermolecular H-bonds
(3) Formation of peptide bonds
(4) The folding of the coiled polypeptide chains

153. Identify the newly synthesised chain given below:



- (1) Peptide (2) Nucleic acid
(3) Lipid (4) Polysaccharide

154. A protein with α -helix or β -helix represents :-

- (1) Secondary structure
(2) Quaternary structure
(3) Tertiary structure
(4) Primary structure

155. Keratin found in hair and nails show a secondary structure called :-

- (1) Parallel sheet
(2) α -helix
(3) Antiparalled sheet
(4) Globular conformation

156. The charged molecule which is electrically neutral is known as :-

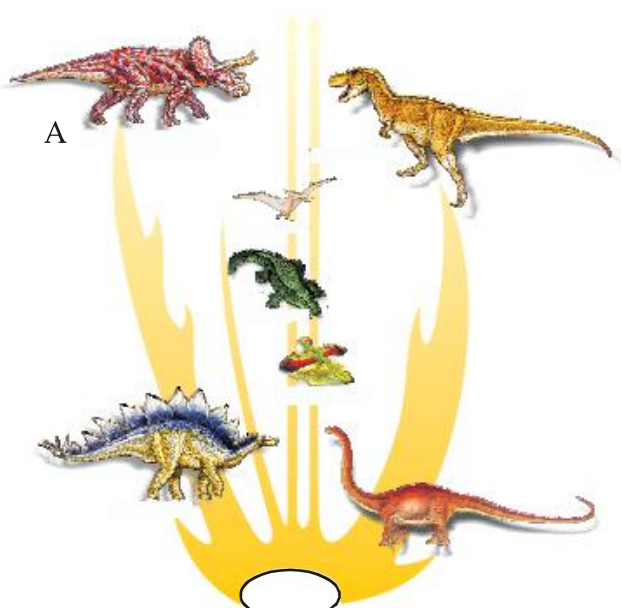
- (1) Amino acid (2) Zwitter ion
(3) Amide (4) Peptide

157. A nucleotide is formed of:-

- (1) Purine, pyrimidine and phosphate
(2) Purine, sugar and phosphate
(3) Nitrogen base, pentose sugar and phosphate
(4) Pyrimidine, sugar and phosphate

158. Building block of nucleic acid is _____:-

- (1) Nucleotide (2) Nucleoside
(3) Amino acid (4) Fatty acid

159. If all the peptide bonds of a protein are broken down, then what would remain ?
 (1) Amino acids (2) Peptides
 (3) Polypeptides (4) Oligopeptides
160. Relation between amino acid and protein is similar to the one found between :-
 (1) Glucose and fructose
 (2) Thymine and uracil
 (3) Nucleosides and nucleic acid
 (4) Nucleotides and nucleic acid
161. Among the following how many matches are correct :-
 (a) Biogenetic law – Haeckel
 (b) Philosophie zoologique – Lamarck
 (c) Use and disuse of organs – Darwin
 (d) Principle of population – Malthus
 (1) Only one statement is correct
 (2) Only two statements are correct
 (3) Only three statements are correct
 (4) All four statements are correct
162. Early earth have which gases :-
 (1) Water vapour and methane
 (2) Carbondioxide
 (3) Ammonia
 (4) All of these
163. Which of the following was the biggest carnivores dianosaur, having 20 feet height, dagger like teeth :-
 (1) Stegosorous
 (2) Brachiosaurus
 (3) Tyrranosaurus
 (4) Triceratops
164. During evolution which of the following was the first amphibian :-
 (1) Lobe fins (2) Synapsids
 (3) Therapsids (4) Shrews
165. Connecting link between apes and human is :-
 (1) Australopithecus
 (2) Shivapithecus
 (3) Ramapithecus
 (4) Dryopithecus
166. Largest cranial capacity was found in :-
 (1) Pecking man
 (2) Neanderthal man
 (3) Java man
 (4) Cro-magnon man
167. Which of the following amino acid was not formed during simulation experiment of Stanley Miller ?
 (1) Alanine (2) Aspartic acid
 (3) Glycine (4) Glutamic acid
168. Which of the following is comparatively a strong evidence of evolution and indicates common origin?
 (1) Eyes of octopus and of mammals
 (2) Potato and sweet potato
 (3) Heart of reptiles, birds and mammals
 (4) Wings of butterfly and crow
169. Identify the animal-A in the family tree of dinosaurs in the adjacent figure ?
- 
- (1) Stegosaurus (2) Pelycosaurs
 (3) Triceratops (4) Brachiosaurs
170. Mark the incorrect statement regarding big bang theory that attempts to explain the origin of universe ?
 (1) It happened about 20 BYA
 (2) Single huge explosion
 (3) Unimaginable in physical terms
 (4) Series of multiple explosion

171. Which of the following is not related with Darwin?

- (1) Population has variations in characteristics.
- (2) Fitness is the end result of the ability to adapt and get selected by nature.
- (3) Evolution is driven by use and disuse of organs.
- (4) Branching descent and natural selection.

172. Which of the following is an evolutionary force that can produce new alleles or can change frequencies of existing alleles in any direction.

- (1) Natural selection
- (2) Genetic drift
- (3) Mutation
- (4) Reproductive isolation

173. Process of evolution of different species in a given area starting from a point and literally radiating to other habitats is called :-

- (1) Adaptive convergence
- (2) Parallel evolution
- (3) Adaptive radiation
- (4) Divergent evolution

174. According to darwin, the founder of theory of natural selection, fitness refers ultimately to :-

- (1) Built in variation
- (2) Reproductive fitness
- (3) Struggle
- (4) Physical fitness

175. In galapagos island, Darwin found different varieties of finches, originated from a common ancestor. The original variety of these finches was:-

- (1) Insectivorous
- (2) Vegetarian
- (3) Seed-eating
- (4) Cactus eating

176. Genetic drift operates rapidly in :-

- (1) Small population
- (2) Large population
- (3) Island population
- (4) Species

177. Match the following columns and find correct combination :-

	Column I		Column II
A	Darwin	P	Mutation theory
B	De vries	Q	Protenoid bodies
C	Pasteur	R	Origin of species
D	Fox	S	Special creation
		T	Swan-necked experiment

- (1) A = R, B = P, C = T, D = Q
- (2) A = P, B = Q, C = R, D = S
- (3) A = T, B = R, C = Q, D = P
- (4) A = R, B = T, C = P, D = Q

178. All of the following arrangements were made by S.L. Miller to create primitive earth like conditions in a laboratory scale, except one :-

- (1) Electric discharge as source of energy
- (2) Closed flask containing CH_4 , H_2 , NH_3 in the ratio of 2 : 2 : 1
- (3) Killed yeasts in a presterilised flask
- (4) Water vapour through boiling water.

179. What will be the age of a fossil recovered from a rock sediment containing 100 gm of radioactive carbon (C^{14}) and 300 gm stable isotope (N^{14}) in a sample taken from the same rock and half life of C^{14} is 5730 years ?

- (1) 11460 years
- (2) 17190 years
- (3) 818 years
- (4) 40110 years

180. Mark the incorrect statement about the first non-cellular form of life ?

- (1) Originated about 3 billion years back
- (2) It was a cluster of nucleoprotein surrounded by lipid coat
- (3) They would have been giant molecules like RNA, protein etc.
- (4) These capsules have molecules with no ability of self replication.